higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PULP AND PAPERMAKING TECHNOLOGY
NQF LEVEL 4

(6050394)

1 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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SECTION A

QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK. Correct the statement if it is FALSE.

1.1 A horizontal pulper is mostly used as an under-machine broke pulper.

1.2 Sheet thickness is not influenced by the type of additives added to the furnish.

1.3 Paper is normally tested for process control, quality control and product development.

1.4 Evaporating spent liquor increases the solid content of invaluable components present in the liquor.

1.5 Slaking is the mixing of green liquor with lime.

1.6 Fibrillation takes place during pulping.

1.7 The couch roll improves web formation by introducing shear in the mix.

1.8 Evaporation drying rate is used to determine dryer performance.

1.9 Sulfite cooking liquor is a mixture of H_2SO_3 and HSO_3^-.

1.10 A waste-heat boiler produces steam that can be used at other parts of the pulp and paper mill.

(10 × 1) **[10]**

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QUESTION 2

Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–J) next to the question number (2.1–2.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	Mechanical force	A	can be recycled for re-use
2.2	Refining	B	mixing of green liquor with lime to form a slurry
2.3	Fluid shear	C	material that can withstand extremely high temperatures
2.4	Yankee	D	pulls apart fibres during pulping
2.5	Web control	E	used to dry low-weight paper products
2.6	Black liquor	F	mechanically conditions the pulp fibres for pulping to improve pulp properties
2.7	Refractory	G	combusted in a recovery furnace
2.8	Heavy black liquor	H	important property of wound paper
2.9	Slaking	I	force experienced by an object when a fluid flows over it
2.10	Property of black liquor	J	dry matter content

(10 × 1) **[10]**

QUESTION 3

Define the following papermaking terms:

3.1 Blow box

3.2 Fluidisation

3.3 Wet strength

3.4 Calorific value

3.5 Evaporator economy

(5 × 2) **[10]**

TOTAL SECTION A: 30

SECTION B

QUESTION 4

4.1 4.1.1 What is the difference between *low-consistency* and *high-consistency* pulpers? (4)

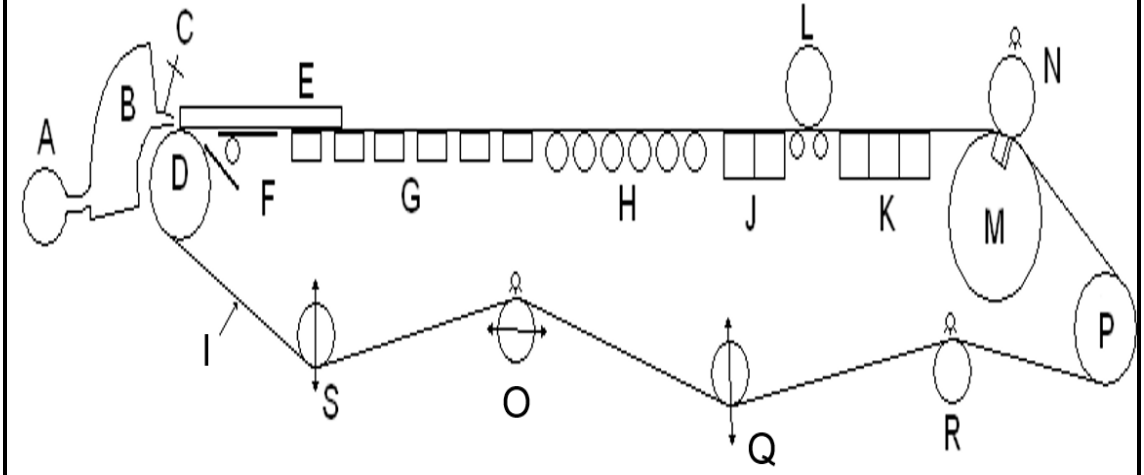
4.1.2 What are the THREE main components of the pulper? (3)

4.1.3 Name TWO other types of pulpers used in the paper industry. (2)

4.2 Discuss the principle of operation of the reverse cleaner. (3)
[12]

QUESTION 5

Study the following diagram of a section of a Fourdrinier machine and answer the questions.



5.1 Name the parts labelled A to S. Write only the answer next to the letter (A–S) in the ANSWER BOOK. (9)

5.2 What are the functions of the parts labelled J, N and P? (3)

5.3 What happens when the sheet reaches point K with a moisture content that is too high? (2)
[14]

QUESTION 6

6.1 Give a short description of how conduction drying occurs. (4)

6.2 List ONE corrective action for each of the following winder defects:

6.2.1 Paper profiles

6.2.2 Core slippage

6.2.3 Slack start

6.2.4 Roll alignment

(4 × 1) (4)

6.3 List FIVE parameters that affect winder quality. (5)

[13]

QUESTION 7

7.1 What are the advantages of black liquor oxidation? (5)

7.2 7.2.1 What is meant by the *heating value* of spent liquors? (1)

7.2.2 When does black liquor have a high heating value? (2)

7.3 Frothing can be a problem during black liquor oxidation in evaporators. The use of defoamers is one way to combat frothing.

Suggest another way to fight this problem and explain how it works. (4)

7.4 7.4.1 List THREE important equipment items used during the recovery of spent liquor with the Copeland process. (3)

7.4.2 Of what material should these equipment items be constructed for the NSSC process and why? (2)

[17]

QUESTION 8

8.1 What is the purpose of a causticising plant? (2)

8.2 Give the names of the equipment items used for white liquor preparation and lime recovery. (7)

8.3 Why is calcium bisulphite not recovered? (3)

8.4 Sulphur is burnt in the presence of air in a sulphur burner to produce sulphur dioxide gas.

Give the chemical equation representing this reaction. (2)

[14]

TOTAL SECTION B: 70
GRAND TOTAL: 100

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